

DEPARTMENT OF MATHEMATICS
UNIVERSITY OF MARYLAND
GRADUATE WRITTEN EXAM

August 2019

ALGEBRA (Ph.D. Version)

Instructions to the student

- a. Answer all six questions; each will be assigned a grade from 0 to 10.
 - b. Use a different booklet for each question. Write the problem number and your **code number (not your name)** on the outside of the booklet.
 - c. Keep scratch work on separate pages in the same booklet.
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1. (a) Suppose that A is a subgroup of a group B with $[B : A] = 2$. Let C be a subgroup of B of odd order. Show that $C \subseteq A$.

- (b) Let G be a finite group and suppose that there exist subgroups

$$G = G_0 \supset G_1 \supset G_2 \supset \cdots \supset G_j = H$$

with $[G_i : G_{i+1}] = 2$ for all i . Suppose H has odd order. Show that H is normal in G .

- (c) Let G be a group of order $2^j k$, where k is odd. Suppose G contains a normal subgroup H of order k . Show that there exist subgroups

$$G = G_0 \supset G_1 \supset G_2 \supset \cdots \supset G_j = H$$

with $[G_i : G_{i+1}] = 2$ for all i .

2. Let $B(x, y)$ be a bilinear form on $\mathbb{R}^n$ (so $B : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$, and $B(ax_1 + bx_2, y) = aB(x_1, y) + bB(x_2, y)$ for $a, b \in \mathbb{R}$ and $x_1, x_2 \in \mathbb{R}^n$, and similarly for the second variable).

- (a) Show that there exists a unique linear transformation $A : \mathbb{R}^n \rightarrow \mathbb{R}^n$ such that $B(x, y) = \langle Ax, y \rangle$ for all $x, y \in \mathbb{R}^n$, where $\langle, \rangle$ is the usual inner product on $\mathbb{R}^n$. (Don't forget to show both existence and uniqueness of A .)

- (b) Let $\{b_1, \dots, b_n\}$ be an orthonormal basis of $\mathbb{R}^n$ (that is, $\langle b_i, b_j \rangle = 0$ if $i \neq j$ and $= 1$ if $i = j$). Show that the sum $\sum_{i=1}^n B(b_i, b_i)$ is independent of the choice of orthonormal basis $\{b_i\}$.

3. Let X and Y be variables. Find all prime ideals of $\mathbb{Z}[X, Y]/(X^2, Y^2)$, where (X^2, Y^2) is the ideal of $\mathbb{Z}[X, Y]$ generated by X^2 and Y^2 .

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4. (a) Let R be a commutative ring with identity and let M be a Noetherian R -module. Let $\phi : M \rightarrow M$ be a surjective homomorphism. Show that ϕ must also be injective. (*Hint:* Look at $\text{Ker}(\phi^n)$.)
 (b) Give an example to show that the hypothesis that M is Noetherian cannot be removed in part (a).
5. (a) Let S_4 be the group of permutations of 4 objects and let A_4 be the subgroup of even permutations. Show that the only subgroup $H \subset S_4$ of order 12 is A_4 .
 (b) Let $f(X) \in \mathbb{Q}[X]$ be an irreducible polynomial of degree 4 and let F be its splitting field. Assume that the Galois group $\text{Gal}(F/\mathbb{Q})$ is S_4 . Let α be a root of $f(X)$ and let $K = \mathbb{Q}(\alpha)$. Show that $\text{Gal}(F/K)$ contains a 2-cycle.
 (c) In the situation of part (b), show that there is no field $L \subset K$ with $[L : \mathbb{Q}] = 2$.
6. Let G be a finite group and let $\rho : G \rightarrow GL_2(\mathbb{C})$ be an irreducible 2-dimensional representation of G . Let H be a normal subgroup of G and suppose that the restriction of ρ to H decomposes as the sum of two one-dimensional representations: $\rho|_H = \chi_1 + \chi_2$, where each χ_i is a homomorphism from G to $\mathbb{C}^\times$. Let $0 \neq v_1, v_2 \in \mathbb{C}^2$ be linearly independent vectors in $\mathbb{C}^2$ such that $\rho(h)v_i = \chi_i(h)v_i$ for all $h \in H$.
 (a) Let $0 \neq v \in \mathbb{C}^2$. Show that there exists $g \in G$ such that v and $\rho(g)v$ are linearly independent.
 (b) Let $g \in G$ and $h \in H$. Let $w_1 = \rho(g)v_1$. Show that $\rho(h)w_1 = \chi_1(g^{-1}hg)w_1$.
 (c) Let g be as in part (a) for $v = v_1$ (so v_1 and $\rho(g)v_1$ are linearly independent). Show that $\chi_2(h) = \chi_1(g^{-1}hg)$ for all $h \in H$. (*Hint:* Write $w_1 = av_1 + bv_2$.)